# The Climate Credit Channel: Physical Shocks and Monetary Policy Transmission in South Africa

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# Monetary policy transmission: beyond the interest rate channel

- Traditional interest rate mechanism: policy rate $\rightarrow$ real rate $\rightarrow$ cost of capital $\rightarrow$ investment (Mishkin 1995)
- Empirical puzzles that this mechanism struggles with (Bernanke and Gertler 1995):
  - Spending responses are too large and too fast relative to estimated cost-of-capital elasticities
  - Heterogeneity: firms respond differently to the same policy shock, particularly depending on their financial position and access to credit.
- The credit channel is not a separate channel: it interacts with the conventional interest rate effect through the **external finance premium**

# The external finance premium (EFP)

**Definition.** The wedge between the cost of funds raised externally (debt) and the opportunity cost of internal funds:

$$\text{EFP}_t \equiv r_t^{ext} - r_t^{int} > 0$$

- Why is it positive? Asymmetric information, agency costs, costly verification and monitoring (Bernanke and Gertler 1989; Hubbard 1998)
- The external finance premium falls when borrowers have stronger balance sheets and more collateral relative to their assets, because lenders see them as less risky (Bernanke et al. 1996)

# Lending rates: the price of credit

- Textbook pass-through from the policy rate $r^p$ to bank lending rates $r^L$ is complete and fast:

$$r^L_{b,t} \;=\; \alpha_b + \beta_b\, r^p_t + \varepsilon_{b,t}$$

- Full passthrough case, $\beta_b = 1$
- But in practice pass-through is incomplete, sluggish and heterogeneous across banks, and depends on bank balance sheets (Altavilla et al. 2020)
- Pass-through is faster when the policy move is anticipated (Kleimeier and Sander 2006)
- Lending rates embed the premium: $r^L = r^p + \text{EFP}$

# The Credit Channel

- Financial frictions interact with monetary policy transmission through two channels:
- **(Borrower) Balance sheet channel:** policy affects borrowers' net worth, cash flow and collateral.
  - A policy tightening weakens borrower balance sheets through three routes (Bernanke and Gertler 1995; Kiyotaki and Moore 1997):
    - **Cash-flow effect:** higher debt service burden
    - **Collateral effect:** lower asset prices reduce the value of collateral
    - **Demand effect:** lower sales and revenues
- Weaker balance sheets $\rightarrow$ higher EFP $\rightarrow$ lower spending, with a "flight to quality" away from weak borrowers (Bernanke et al. 1996; Gertler and Gilchrist 1994)
- The same logic applies to households via wealth and debt service burdens (Slacalek et al. 2020)
- **Bank lending channel:** policy affects banks' capacity or willingness to supply loans.
- Monetary tightening can reduce bank funding and constrain loan supply.
- Strongest for small, illiquid banks (Kashyap and Stein 2000); composition of external finance shifts (Kashyap et al. 1993)
- In South Africa the bank lending channel is operative (Pirozhkova and Viegi 2025)

## The credit channel is not fully understood

- The **magnitude** of the EFP response to policy varies across time, borrowers and products; even its originators flagged that its quantitative importance is unsettled (Bernanke and Gertler 1995)
- **Who responds?** Borrowers' Leverage (Ottonello and Winberry 2020) versus other firm characteristics eg. age and dividend status (Cloyne et al. 2023) give different answers about which balance sheets matter
- **When?** Transmission is weaker in recessions (Tenreyro and Thwaites 2016); low rates induce bank risk-taking (Jiménez et al. 2014)
- **Pass-through heterogeneity:** Lenders balance sheets shape how much of a policy move reaches borrowers (Altavilla et al. 2020)
- **Geography** The bulk of the evidence is from the United States and the euro area. Transmission in emerging and low-income economies is weaker and less reliable, reflecting shallower financial markets and bank concentration (Mishra et al. 2012)
- **Identification:** policy announcements convey information about the outlook, contaminating measured responses (Nakamura and Steinsson 2018; Jarociński and Karadi 2020)

## Why surprises rather than the policy rate

- The repo rate $r_t$ is **endogenous**: it responds to inflation, output and, in a climate-exposed economy, potentially to food prices and the weather
- Regressing credit on $r_t$ confounds policy with the conditions that prompted policy
- We need variation in policy that is orthogonal to the policymaker's and the market's information sets (Romer and Romer 2004)
- Two families of surprise measures:
  - **Structural / forecast based:** residual of the rate change on forecasts (Romer and Romer; Miyajima)
  - **High frequency / market based:** asset price moves in a narrow window around announcements (Kuttner 2001; Gürkaynak et al. 2005)

# Surprises do not transmit like textbook policy

- Kuttner (2001): the **unanticipated** component of a rate change moves market rates

$$\Delta i_t = \alpha + \beta_e \, \Delta r_t^e + \beta_u \, \Delta r_t^u + \eta_t, \qquad \hat{\beta}_e \approx 0, \;\; \hat{\beta}_u \gg 0$$

- Credit costs respond far more to surprises than a pure expectations model predicts (Gertler and Karadi 2015)
- Announcements also carry information about the outlook, so some "tightening" surprises can actually lower spreads (Nakamura and Steinsson 2018; Jarociński and Karadi 2020)
- Lending-rate pass-through is slower when a move is **unexpected** (Kleimeier and Sander 2006)
- An unanticipated move is exactly where EFP frictions have the most room to bite, which is part of why we focus on surprises

## Climate and the macroeconomy

- Temperature shocks lower growth, mainly in poorer countries (Dell et al. 2012)
- Disasters depress output and recovery is slow in disaster-prone economies (Noy 2009; Felbermayr and Gröschl 2014; Cantelmo et al. 2023)
- Mechanisms map directly onto the EFP: production losses, capital damage, lower collateral values, lower net worth (Dafermos et al. 2018)

## Climate shocks and prices: the monetary policy nexus

- Extreme weather raises consumer prices, especially food, in developing countries (Heinen et al. 2019); disasters are inflationary in the short run (Parker 2018)
- Temperature shocks raise inflation (Mukherjee and Ouattara 2021), with asymmetric and seasonal effects (Ciccarelli et al. 2024) and persistence of about 12 months (Kotz et al. 2024)
- Central bank dilemma: a climate shock is a supply shock but can generate second-round effects and expectations risk.
- Sector specific inflation pressure from climate shocks in SA (Merrino and Sibande 2026)
  - Optimal policy under relative price shocks (Aoki 2001); EM central banks may need to react to food inflation (Anand et al. 2015)

## Climate shocks, banks and credit

- After disasters, banks reallocate credit toward affected areas (Cortés and Strahan 2017), engage in recovery lending (Koetter et al. 2020), and adjust risk exposure (Schüwer et al. 2019)
- Firms hit by adverse weather draw on bank credit lines within a quarter (Brown et al. 2021)
- Banks reprice risk: loan spreads rise even for borrowers at risk but not directly affected (Correa et al. 2026)
- Households: insurance payouts are used to repay mortgages (Gallagher and Hartley 2017); disasters raise financial distress (Ho et al. 2023); mortgage standards tighten in disaster-prone areas (Duanmu et al. 2022)
- Climate risk is a financial stability issue (Dafermos et al. 2018; Giglio et al. 2021; Hong et al. 2020)

## Conceptual framework (i): corporates

Following the external finance premium logic (Bernanke et al. 1996; Laumer and Schaffer 2025):

$$R_{it}^{c} = R_t + f(D_t/E_t)\,, \qquad f' > 0$$

- A climate shock disrupts supply chains, damages capital and destroys output: revenues and profits fall
- Collateral values fall; debt $D_t$ rises; equity $E_t$ falls $\Rightarrow$ $f(D_t/E_t)$ rises
- **Higher Rate:** the climate shock raises the premium at the same time as a tightening raises $R_t$
- **Same Rate:** firms with cash buffers or insurance see no change in $D_t/E_t$ (Sharpe and Suarez 2020)
- **Adaptation-Driven Borrowing:** Another, related transmission avenue, is firms that must rebuild or adapt borrow even at higher rates (Xu et al. 2025)

# Conceptual framework (ii): households

$$R_{it}^{h} = R_t + \phi_t, \qquad \phi_t > 0$$

- $\phi_t$: household-specific spread reflecting creditworthiness, indebtedness, income and collateral; transmission differs across households by income exposure (Slacalek et al. 2020)
- **Higher Rate:** lower income and wealth and damaged collateral raise $\phi_t$; affected households pay more for a given policy rate
- **Same Rate:** liquid assets and insurance absorb losses without widening $\phi_t$; insured households use payouts to repay rather than borrow (Gallagher and Hartley 2017)

# Conceptual framework (iii): banks

- Banks connect the two: climate episodes raise the weight lenders put on borrower balance sheet strength due to increased arrears and financial distress (Ho et al. 2023; Duanmu et al. 2022; Correa et al. 2026)
- Weaker loan demand after a shock can instead leave banks with less scope to reprice, dampening the rate response
- The net effect is an empirical question
- We test the net effect without attributing it to a single mechanism

# The gap this paper addresses

- **Literature A:** monetary policy → credit through the EFP, identified with policy surprises
- **Literature B:** climate shocks → balance sheets, loan pricing, inflation
- **Gap:** The interaction between physical climate shocks and monetary policy transmission to credit markets remains underexplored, particularly in emerging economies.
- Our contribution:
  - Does climate change the transmission of monetary policy to credit markets in South Africa, and how do physical climate shocks affect the strength and persistence of this transmission across credit products and borrowers?
    - Bank-level, product-level lending rates and volumes for South Africa, 2013 to 2025
    - Three identified policy surprises × two population-weighted climate shocks
    - Fixed effects, panel local projections and asymmetric (extreme-state) local projections

## Structural surprises (i): the Romer and Romer measure

Following Romer and Romer (2004), adapted to South Africa by Merrino (2022):

$$\Delta r_t = \alpha + \tilde{r}_t + \sum_{j=0}^{2} \beta_j \Delta \tilde{y}_{t+j} + \sum_{j=0}^{2} \delta_j \tilde{\pi}_{t+j} + \varepsilon_t$$

- $\Delta r_t$: change in the repo rate; $\tilde{r}_t$, $\tilde{y}_t$, $\tilde{\pi}_t$: forecasts of the repo rate, real GDP growth and inflation
- The residual $\varepsilon_t$ is the component of the policy change not explained by what policymakers already expected: $\text{MPShock}_t^{Romer}$

## Structural surprises (ii): the Miyajima measure

Following Miyajima (2024):

$$FE_t^i = ACT_t^i - EXP_t^i, \qquad i \in \{\text{policy rate}, \pi, y\}$$

$$FE_t^r = \alpha + \beta\, FE_t^\pi + \gamma\, FE_t^y + \epsilon_t$$

- The policy rate forecast error is purged of inflation and growth news
- The residual $\epsilon_t$ is the part of the policy miss that markets could not have anticipated from macro news: $\text{MPShock}_t^{Miyajima}$

## High-frequency surprises

- Measure asset price changes in a narrow window around each MPC announcement (Kuttner 2001; Gürkaynak et al. 2005)
- Because little else happens in the window, the change in financial market prices is attributed to policy news (Nakamura and Steinsson 2018)
- For South Africa, Pirozhkova and Viegi (2025) extract target, forward guidance, central bank information and country risk factors from market prices around MPC dates
- We use the **target factor** as our third surprise measure. The target factor is the component of financial-market movements around MPC announcements that captures unexpected changes in the current policy stance.

# The three surprise series

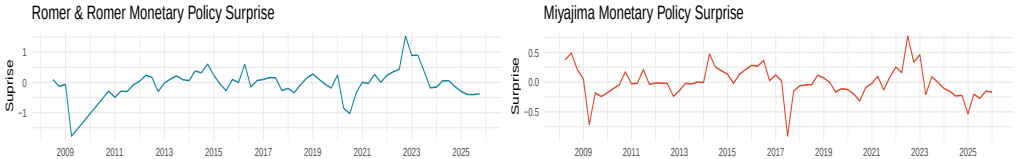



**Figure 1**




**Figure 2**

## Climate shocks as exogenous variation

- We use deviations of population-weighted temperature and precipitation from their calendar-month norms, in standard deviation units:

$$\text{ClimateShock}_t = \frac{x_t - \bar{x}_t}{\sigma_t}$$

- where $x_t$ is climate variable in month $t$, and $\bar{x}_t$ and $\sigma_t$ are the mean and standard deviation of $x_t$ across observations in the same calendar month.
- Weather realisations are plausibly exogenous to credit conditions and to policy within the quarter (Dell et al. 2014; Hsiang 2016)
- Source: ERA5 reanalysis, aggregated over South Africa with population weights

# The climate shocks series

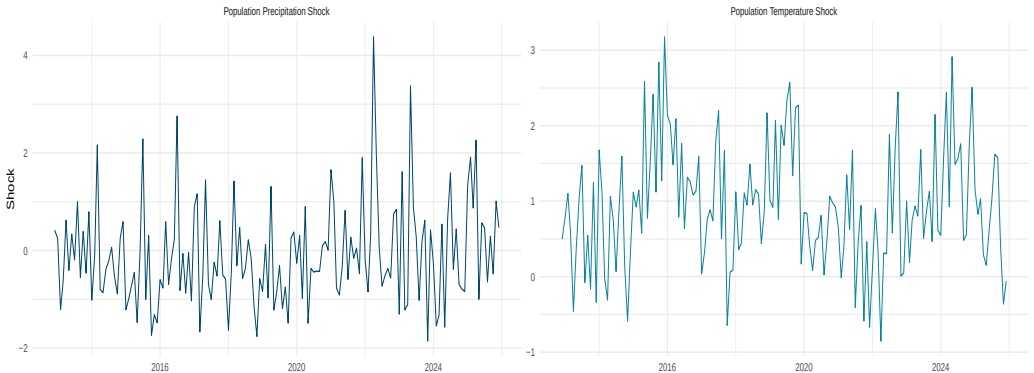



**Figure 3**

Positive values indicate warmer or wetter than the historical monthly norm; negative values cooler or drier. The 2015/16 El Niño drought and the 2022 flooding episodes are visible as tail observations.

# Data

- **Sample:** South Africa, 2013Q1 to 2025Q1, quarterly; bank-quarter panel of the 4 major banking groups
- **Credit (SARB BA900 and BA930 returns, bank level):** lending rates (percentage points) and log credit volumes for corporate and household borrowers, each split into unsecured, secured and mortgage credit, constructed following Sibande et al. (2025) and Sibande and Milne (2025)
- **Monetary policy surprises:** Romer and Romer (SARB forecasts), Miyajima (Bloomberg consensus forecast errors), Target factor (Pirozhkova and Viegi 2025); all aggregated to quarters
- **Climate shocks:** ERA5 population-weighted temperature and precipitation deviations from calendar-month norms
- **Controls (robustness):** GDP growth, USD/ZAR, unemployment rate

# Contemporaneous specification: panel fixed effects

$$\text{Credit}_{s,t} = \beta_0 + \beta_1 \, \text{CS}_t^k + \beta_2 \, \text{MP}_t + \beta_3 \, (\text{CS}_t^k \times \text{MP}_t) + \Gamma X_t + \gamma_s + \tau t + \varepsilon_{s,t}$$

- $\text{CS}_t^k$: climate shock $k$ (temperature or precipitation); $\text{MP}_t$: monetary policy surprise
- $\text{Credit}_{s,t}$: lending rate or log volume for bank $s$ and a given product
- $\gamma_s$: bank fixed effects; $\tau t$: linear trend
- $\beta_2$: effect of a surprise tightening without a climate shock
- $\beta_3$: **the coefficient of interest.** $\beta_3 \neq 0$ means the climate state changes the strength of transmission; the sign relative to $\beta_2$ tells us amplification versus dampening
- Standard errors clustered at the bank level

# Dynamic specification: panel local projections

For each horizon $h = 0, 1, \ldots, 12$ quarters (Jordà 2005):

$$\begin{aligned}\text{Credit}_{s,t+h} - \text{Credit}_{s,t-1} = \beta_0^h + \beta_1^h \, \text{CS}_t^k + \beta_2^h \, \text{MP}_t + \beta_3^h \, (\text{CS}_t^k \times \text{MP}_t) \\ + \Gamma X_t + \gamma_s + \tau t + \varepsilon_{s,t}^h\end{aligned}$$

- Impulse responses: $\widehat{\beta}_2^h$ (policy alone, blue) and $\widehat{\beta}_2^h + \widehat{\beta}_3^h$ (policy with a one standard deviation climate shock, red)
- The vertical gap between the two paths at each horizon is the climate amplification or dampening of transmission
- Driscoll-Kraay standard errors for cross-sectional and serial dependence (Driscoll and Kraay 1998); 68% and 90% bands

## Extreme states: asymmetric local projections

- Average interactions can mask what happens in the tails, which is where the conceptual framework predicts the largest balance sheet damage
- We evaluate the dynamic response at **extreme climate states**:

$$\text{LP-IRF}^h(c) \;=\; \hat{\beta}_2^h + \hat{\beta}_3^h \cdot c, \qquad c \in \{+3, -3\}$$

  - $c = +3$: extreme heat
  - $c = -3$: extreme lack of precipitation

# Contemporaneous evidence: lending rates

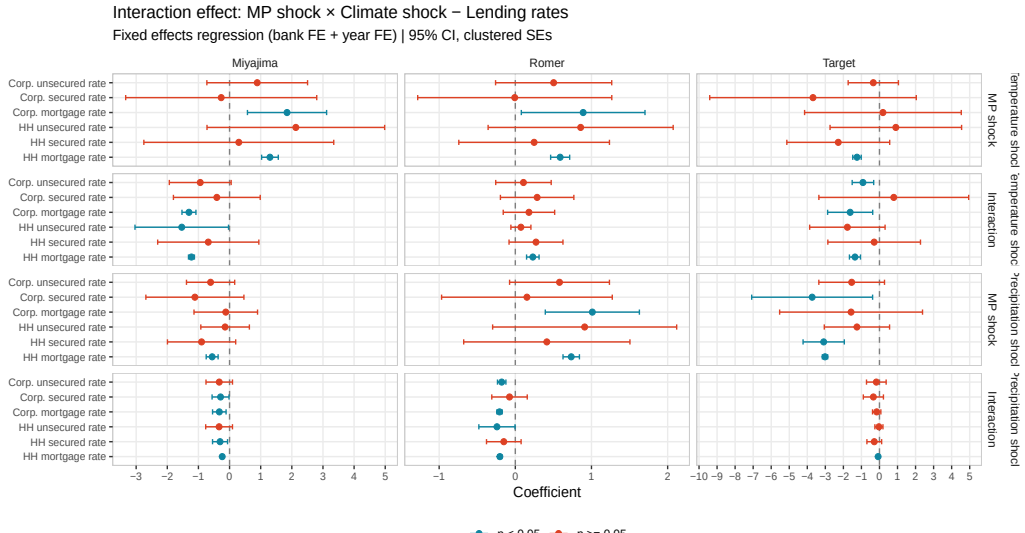



**Figure 4**

# Contemporaneous evidence: credit volumes

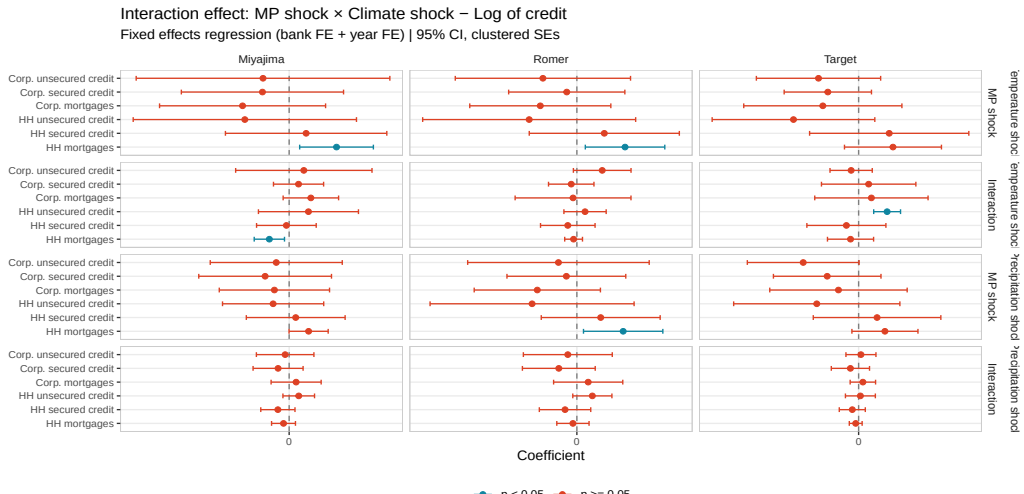



**Figure 5**

# Corporate lending rates: temperature shock interaction

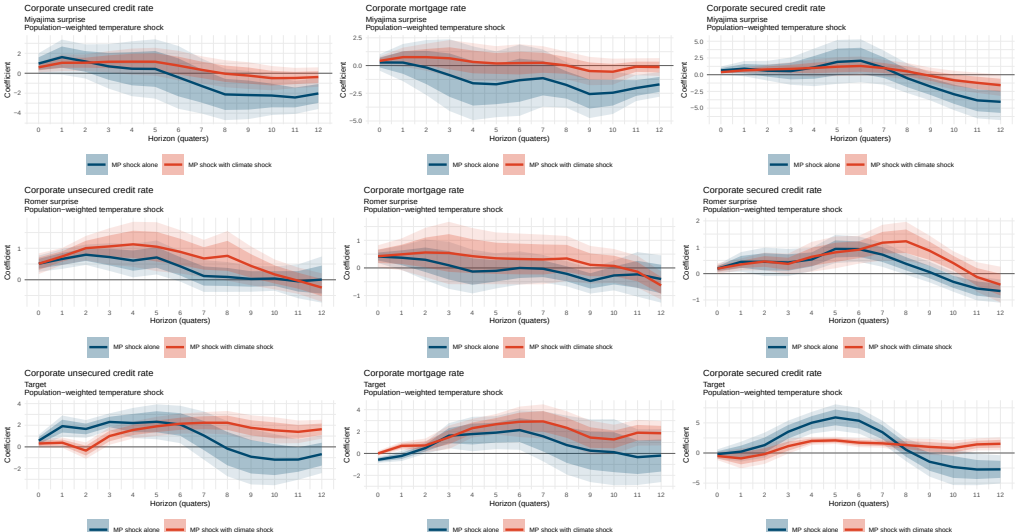



**Figure 6**

# Household lending rates: temperature shock interaction

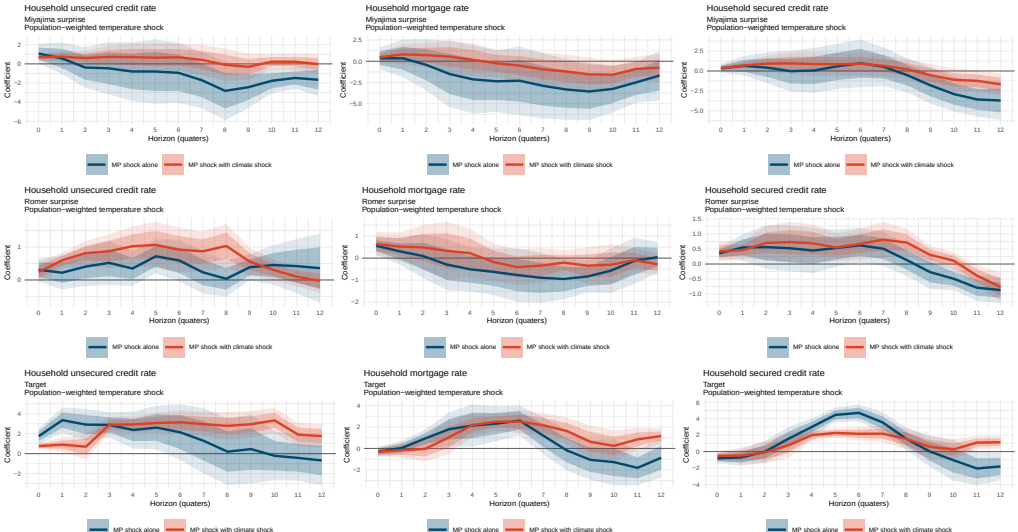



**Figure 7**

# Corporate lending rates: precipitation shock interaction

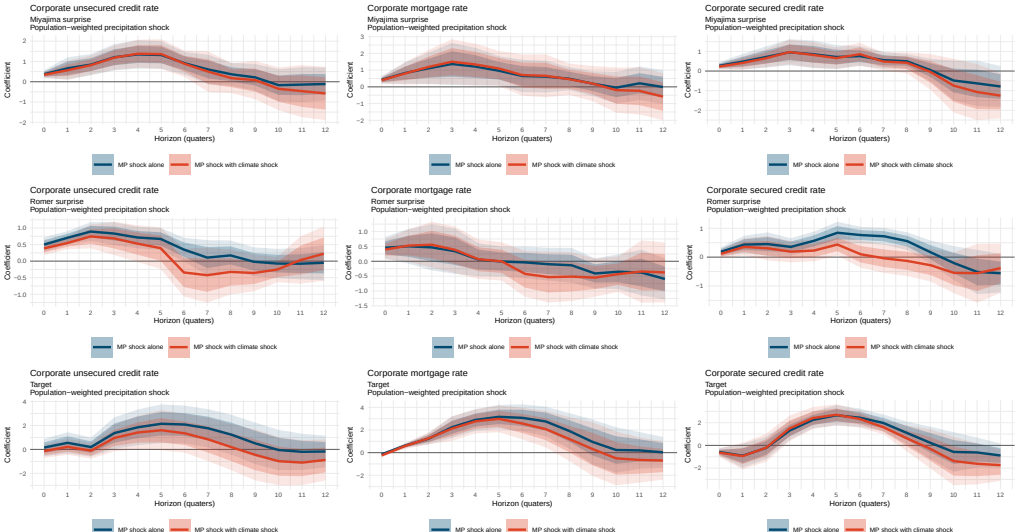



**Figure 8**

# Household lending rates: precipitation shock interaction

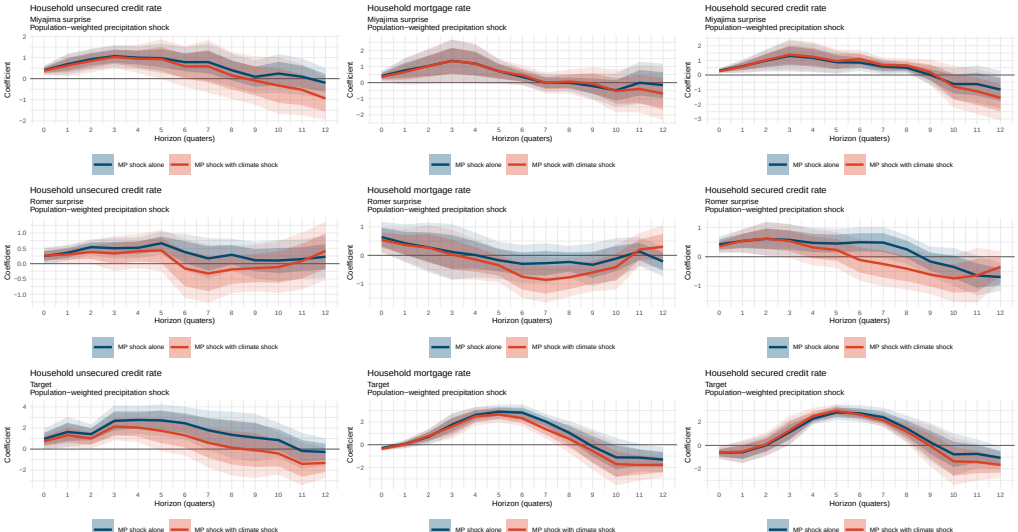



**Figure 9**

# A real-world episode: the 2015/16 El Niño drought in South Africa

| MPC date | Repo decision | Cum. repo | Household mortgage rate | Cum. change | Drought-related rationale |
|---|---|---|---|---|---|
| Nov 2015 | +25 bp to 6.25% | +25 bp | 9.11% (2015Q4) | +23 bp | "worsening drought conditions and their likely impact on food prices" |
| Jan 2016 | +50 bp to 6.75% | +75 bp | 9.72% (2016Q1) | +84 bp | "Impact of the worsening drought on food prices increasingly evident" |
| Mar 2016 | +25 bp to 7.00% | +100 bp | 10.00% (2016Q2) | +112 bp | "Food price pressures, driven by the drought... have intensified by more than previously forecast" |
| Mar 2016 to Jul 2017 | Held at 7.00% | +100 bp | 10.12% (2017Q2) | +124 bp | Lending rate still rising five quarters after the last hike |
| Jul 2017 | Cut 25 bp to 6.75% | **+75 bp** | 9.92% (2017Q3) | **+104 bp** | First cut as food inflation eased; lending rate fell by less than repo |

Source: SARB. Changes relative to 2015Q3 (8.88%)

# Extreme heat: corporate lending rates

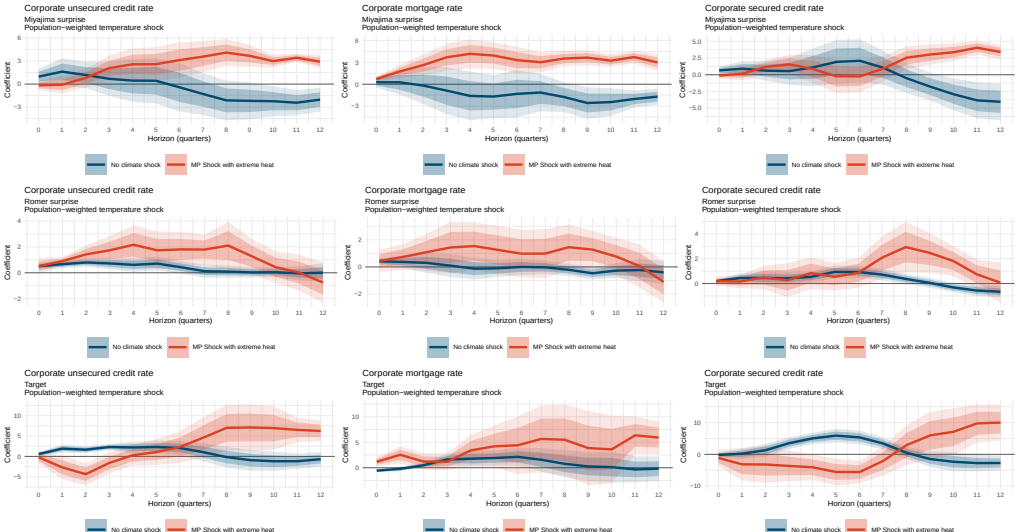



**Figure 10**

# Extreme heat: household lending rates

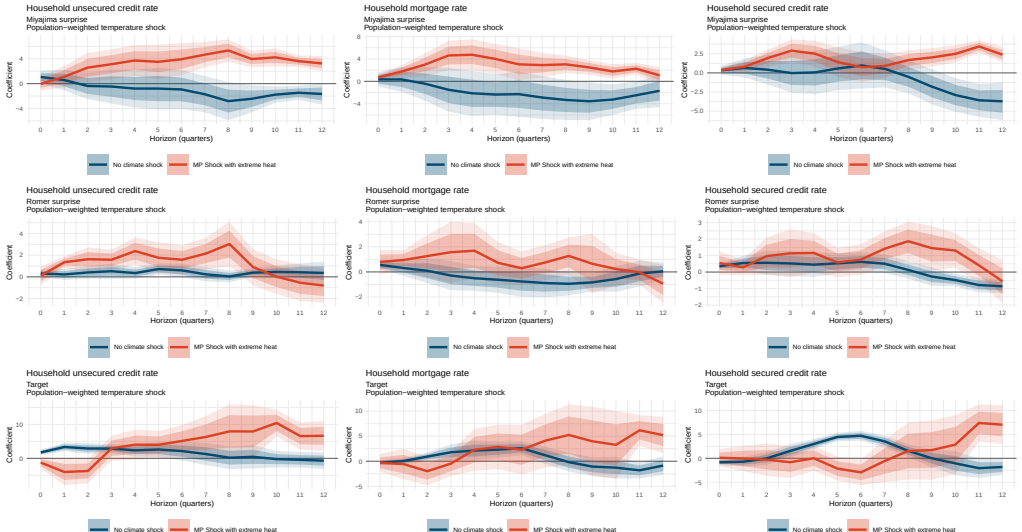



**Figure 11**

# Extreme lack of precipitation: corporate lending rates

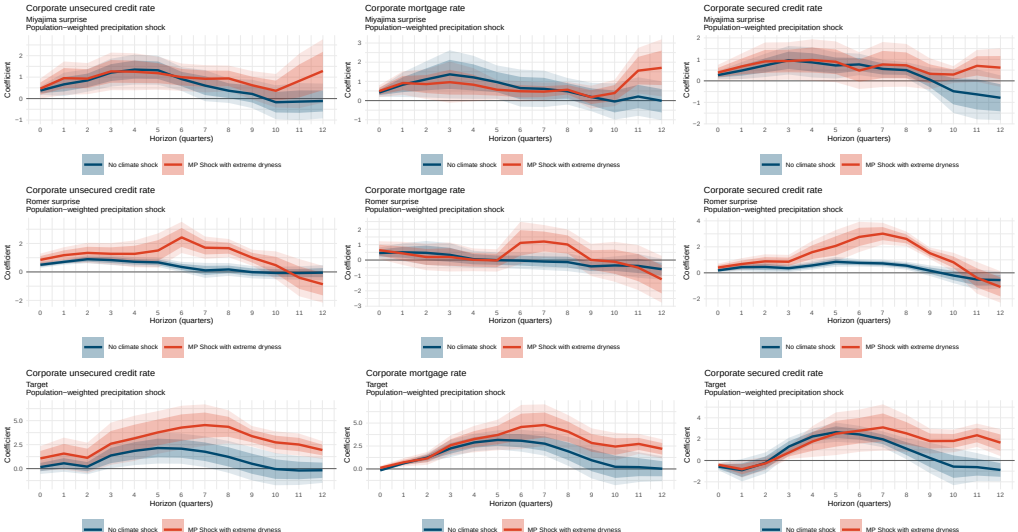



**Figure 12**

# Extreme lack of precipitation: household lending rates

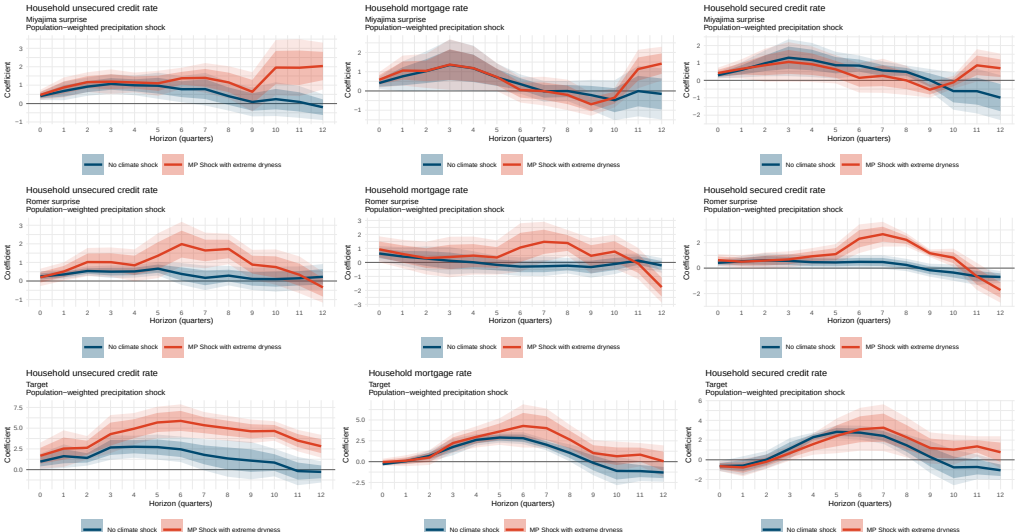



**Figure 13**

# Persistence and timing of the interaction

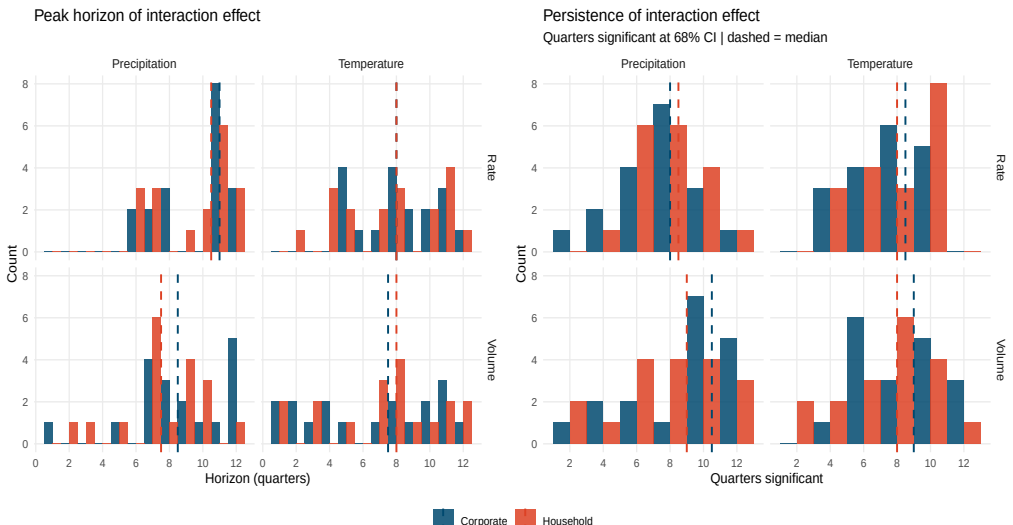



**Figure 14**

## Summary of findings

1. **Transmission is augmented by climate shocks.** The interaction between policy surprises and climate shocks is significant across a large share of product-horizon combinations
2. **The evidence is in prices, not quantities.** Significant interactions concentrate in lending rates; log volumes show little consistent interaction. This is risk repricing through the external finance premium, not rationing
3. **Temperature amplifies; precipitation depends on the threshold.** Heat keeps lending rates higher for longer; moderate rain does not, "drought-like" conditions convincingly amplify
4. **The effect is a tail phenomenon.** Extreme-state responses diverge sharply and durably from the baseline
5. **It is persistent.** Interaction peaks around 8 quarters and remains significant for roughly two years

## Policy implications

- **Disentangle climate from other shocks.**
- **Unintended restrictiveness.** A tightening that coincides with a severe heat or drought episode transmits more strongly and for longer through lending rates than the policy rate path implies. Models that ignore the climate state will overstate the room for tightening and understate how long the restriction lasts
- **Room for a more targeted response.**
- **Climate-vulnerable central banks** would benefit from incorporating physical climate risk into their transmission models, consistent with the financial-stability view of climate (Dafermos et al. 2018; Giglio et al. 2021)

## The End

Thank you

# Corporate lending rates: lagged temperature shock interaction

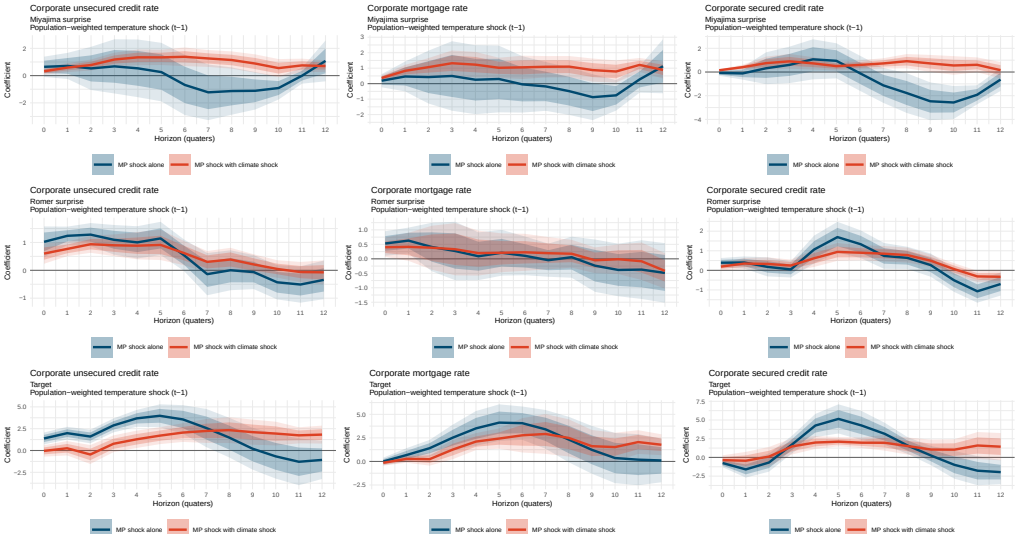



**Figure 15**

# Household lending rates: lagged temperature shock interaction

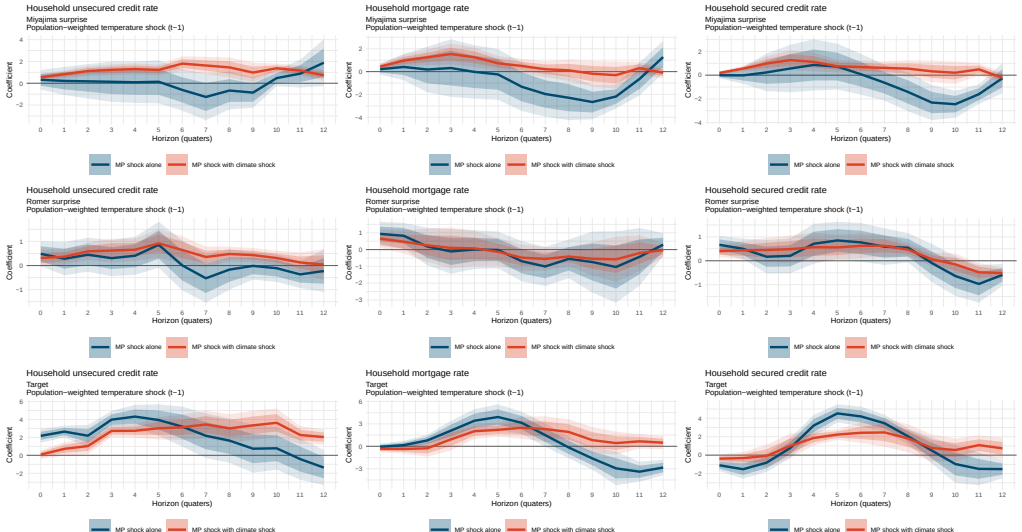



**Figure 16**

# Corporate lending rates: lagged precipitation shock interaction

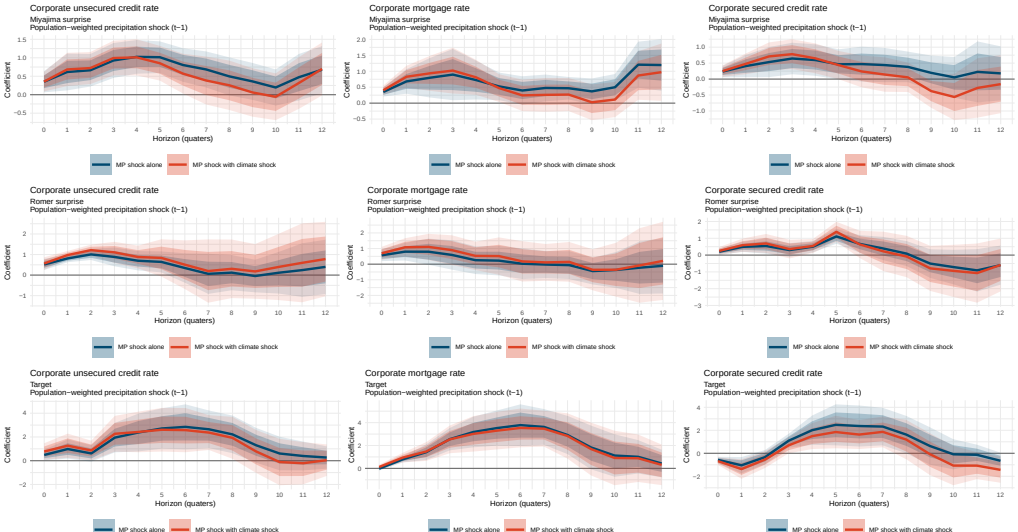



**Figure 17**

# Household lending rates: lagged precipitation shock interaction

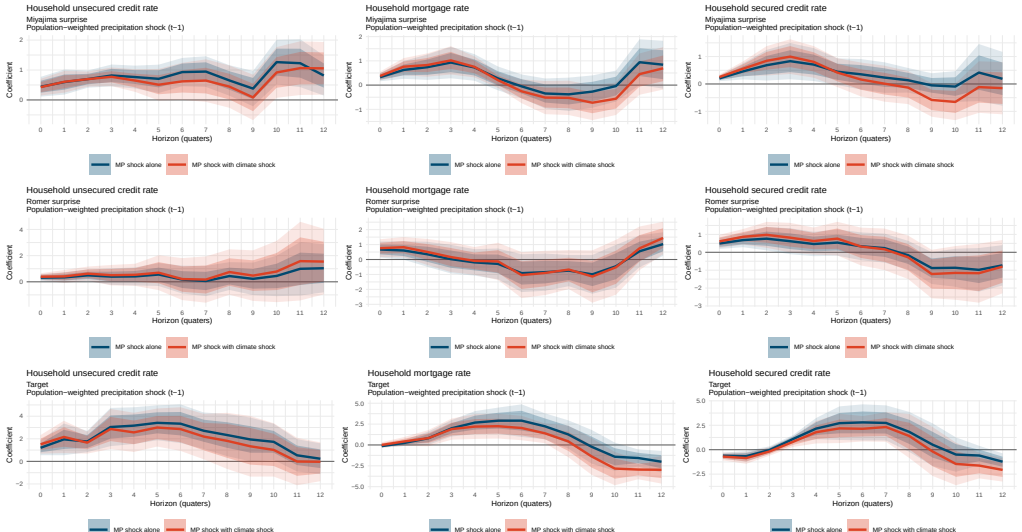



**Figure 18**

# Extreme lagged heat: corporate lending rates

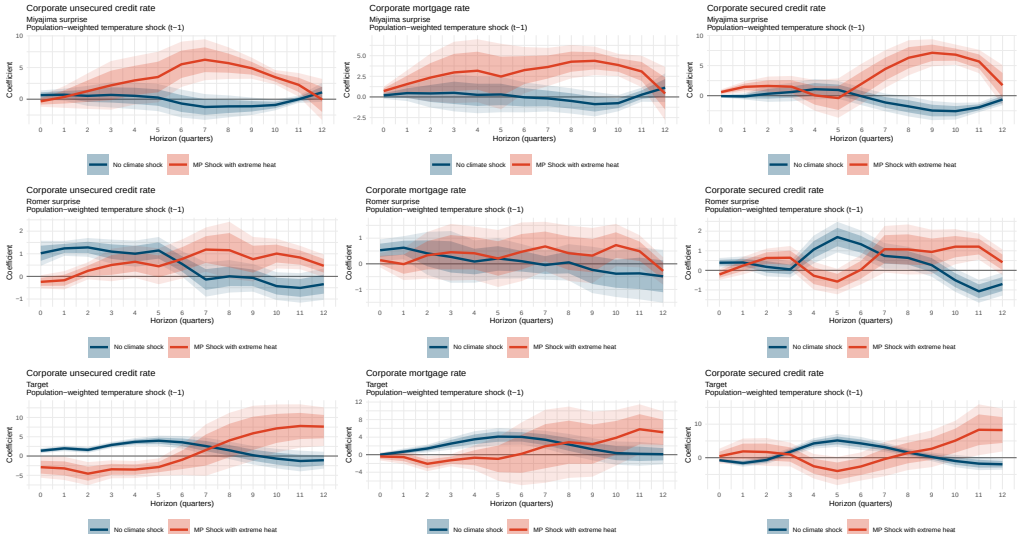



**Figure 19**

# Extreme lagged heat: household lending rates

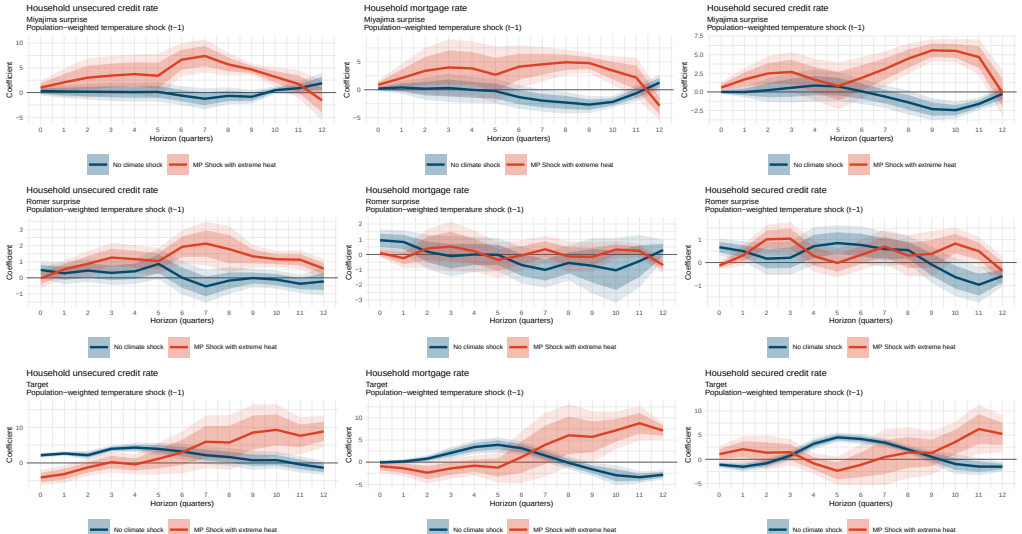



**Figure 20**

# Extreme lagged lack of precipitation: corporate lending rates

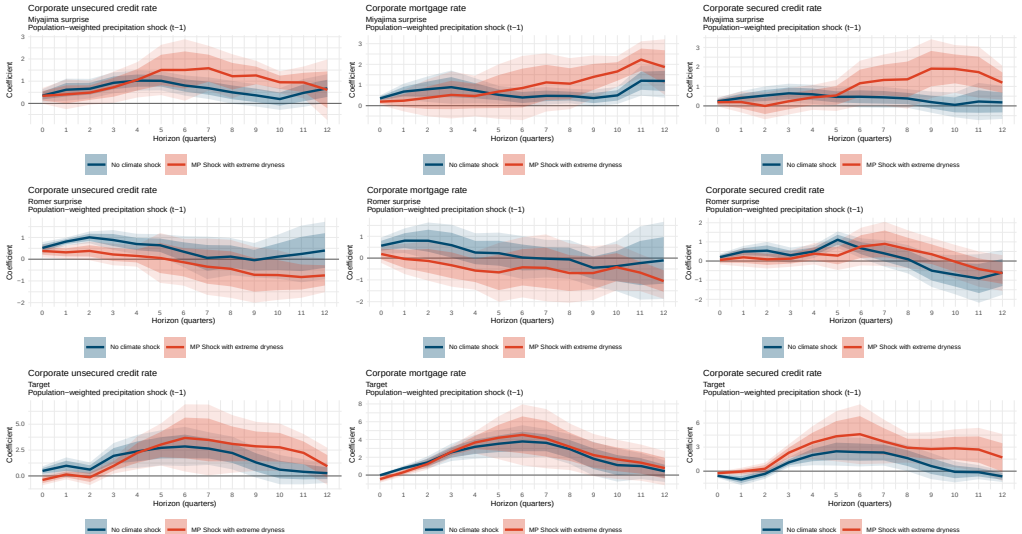



**Figure 21**

# Extreme lagged lack of precipitation: household lending rates

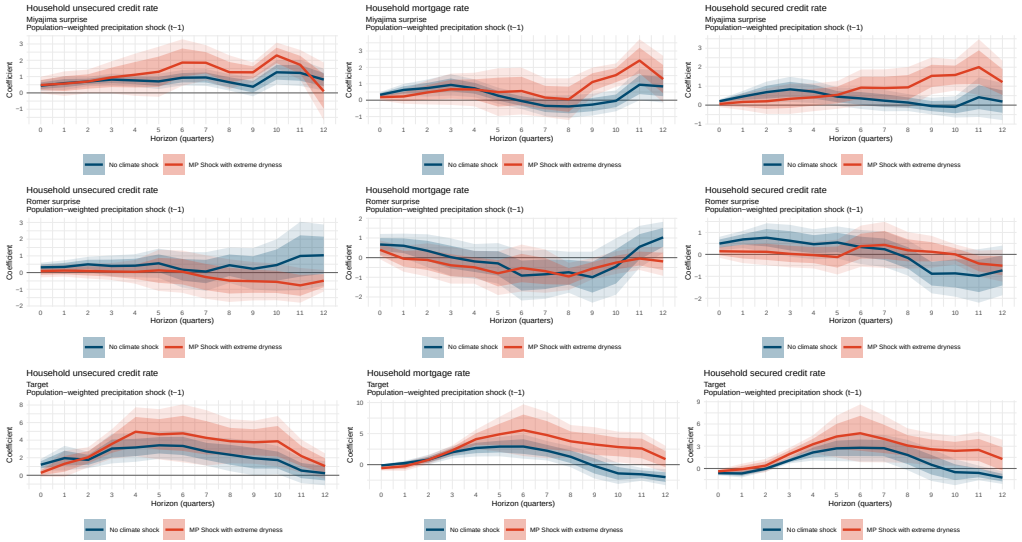



**Figure 22**

# Persistence by credit product

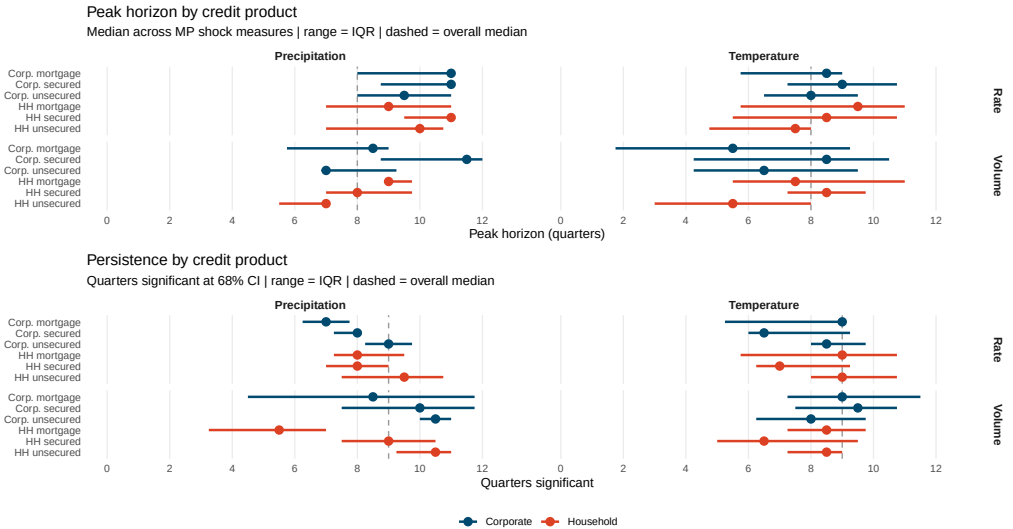



**Figure 23**